RELATIVE ATOMIC MASS

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Because individual atoms are so small, units such as the gram are too large for describing low numbers of atoms. Atomic mass units (amu) are the basic units of mass for atoms. One amu is equal to 1.66×10^{-27} kg.

Chemists need a standard against which they can compare atoms. The first standard was hydrogen (the simplest atom) which gave a hydrogen atom a mass of 1.0000 amu. Later, naturally-occurring oxygen was used instead of hydrogen and assigned a mass of 16.0000 amu. However, oxygen exists as isotopes, where some atoms have a different number of neutrons and hence different masses. In 1961 the isotope carbon–12 (^{12}C) was adopted as the standard measure and one atom of carbon–12 is defined as having a mass of 12 amu. In other words, the mass of 1 amu is one-twelfth of the mass of a carbon–12 atom.

Because the masses of all atoms are compared to this standard, they are often called **relative atomic masses** (*relative* meaning compared to each other). Some older textbooks refer to atomic mass units as **daltons**, named after John Dalton (1766–1844).

Most Periodic Tables give relative atomic masses as well as atomic numbers. Remember:

- **Atomic number** is the number of protons (and is also equal to the number of electrons)
- **Atomic mass** (mass number) is the number of protons plus the number of neutrons
- **Relative atomic masses** of elements are often not whole numbers because in nature they are mixtures of isotopes
- The term **atomic weight** is often used instead of atomic mass.

1 Give the term for the definition: 'When a chemical element has atoms with the same atomic number but with differing atomic mass numbers.'

2 An atom of germanium is 6 times as heavy as an atom of carbon–12.

a Give the relative atomic mass of germanium in amu.

b Give the symbol for germanium.

3 The relative atomic mass of titanium is 48.

a How many times heavier than an atom of carbon–12 is an atom of titanium?

b Give the symbol for titanium.

Compounds are groups of atoms joined together. The formula of a compound can be used to calculate its relative formula mass. Sometimes the term **relative molecular mass** is used instead of relative formula mass. Technically, relative molecular mass should only be used when dealing with covalent compounds.

EXAMPLE

What is the mass of a molecule of carbon dioxide CO_2?
(Note that the relative atomic mass of carbon is 12.0 and of oxygen is 16.0.)

From the formula, there is one carbon atom and two oxygen atoms in a molecule of carbon dioxide.

Mass = $(1 \times 12.0) + (2 \times 16.0) = 12.0 + 32.0 = 44.0$ amu (Don't forget the units!)

4 Find the relative formula mass of each compound. Show your working.

a magnesium oxide, MgO

b iron (III) oxide, Fe_2O_3

c copper (II) sulfate, $CuSO_4$

d ammonium chloride, NH_4Cl.

e sucrose, $C_{12}H_{22}O_{11}$

f potassium carbonate, K_2CO_3

5 Mark each statement true or false.

a The atomic mass unit (amu) is a heavier unit than the gram.

b The atomic mass of an element is the average mass of its isotopes, taking into account the proportion of each isotope in a natural sample.

c Relative atomic masses are based on the mass of a naturally occurring hydrogen atom.

d One amu is equal to the mass of a neutron.

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